

Exercise Sheet for Week 4 - Simply-typed λ -calculus

Assigned: Monday 21 October
Suggested Deadline: Monday 28 October

Questions marked with * are optional, more difficult challenges.

Question 1: Typing closed Church-style terms

For each of the following closed Church-style λ -terms, find its unique typing derivation or prove that it is untypable.

- (a) $\lambda x : \mathbb{B}. x$
- (b) **true**
- (c) $\lambda x : \mathbb{B} \rightarrow \mathbb{B} \rightarrow \mathbb{B}. \lambda y : \mathbb{B}. x y y$
- (d) $\lambda f : \mathbb{B} \rightarrow \mathbb{B}. \lambda x : \mathbb{B}. f(f(f(f(x))))$
- (e) $\lambda x : \mathbb{B} \rightarrow \mathbb{B}. x \lambda y : \mathbb{B}. y$
- (f) $\lambda x : \mathbb{B}. \lambda x : \mathbb{B} \rightarrow \mathbb{B}. x x.$
- (g) $\lambda x : (\mathbb{B} \rightarrow \mathbb{B}) \rightarrow \mathbb{B}. \lambda y : \mathbb{B} \rightarrow \mathbb{B}. x \lambda z : \mathbb{B}. x y.$
- (h) $\lambda z : \mathbb{B}. \text{if } z \text{ then } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. z \text{ else } \lambda f : \mathbb{B} \rightarrow \mathbb{B}. f z$

Question 2: Typing open Church-style terms

For each of the following Church-style λ -terms t , find all typing contexts Γ and types T such that $\Gamma \vdash t : T$ is a valid typing judgement.

- (a) x
- (b) $x x$
- (c) $x y x$
- (d) $x (y x)$
- (e) $x \lambda y : \mathbb{B}. x y.$
- (f) $\lambda y : \mathbb{B}. x (x y).$
- (g) **if** x **then** x **else** $x.$
- (h) $z (y \text{true}) (y x) (x \text{false})$